

Multiple Choice

1. **Answer:** A

Options: A) The set of all finite subsets of $\mathbb{R}$; B) $\mathbb{R}$; C) The set of all infinite subsets of $\mathbb{R}$; D) The set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$; E) The set of all functions from $\{0, 1\}$ to $\mathbb{R}$

Note: Correct option: A - The set of all finite subsets of $\mathbb{R}$

2. **Answer:** A

Options: A) $2/50$; B) $1/25$; C) $1/30$; D) $3/75$; E) $4/100$

Note: Correct option: A - $2/50$

3. **Answer:** A

Options: A) \$1.50; B) \$0.50; C) \$0.20; D) \$0.60; E) \$0.10

Note: Correct option: A - \$1.50

4. **Answer:** A

Options: A) 5; B) 25; C) 7; D) 6; E) 10

Note: Correct option: A - 5

5. **Answer:** A

Options: A) 24; B) 28; C) 20; D) 16; E) 8

Note: Correct option: A - 24

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '66'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3980'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 578 r 2. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 103. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key